

2.5 MW Pulse Power Supply

Team Number: 57

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Executive Summary

The goal of this project is to build a pulsed power modulator prototype that is safe enough to be used in an academic environment in order to fill the lack of hands-on education with PWMs in the college laboratory due to the unsafe nature of high voltage. The prototype is meant to demonstrate how a high voltage pulse power system would behave, with the main elements demonstrated being switching, sensing, and control. With these three components combined, a pulse power train can form. After weeks of rigorous design testing and implementation, the team has achieved this goal and verified results through an oscilloscope, with an LED for easy visual demonstration. The prototype input voltage is 1-12V, with a duty cycle range of 1-99%, and frequency up to 8 kHz that produces about 0.6 W of power per pulse. The budget used for this project is estimated to be \$185, with some components already being owned. With a larger budget, the team would be able to scale up the circuit design with more expensive components to increase the magnitude of output power.

Introduction

Pulsed power systems deliver energy in short, high-intensity bursts, enabling very high instantaneous power from modest average sources. They are essential to applications such as particle accelerators, radar, medical devices, and advanced industrial systems. However, full-scale pulsed-power modulators are unsafe and impractical for student laboratories, creating a gap between academic theory and real-world systems.

A low-voltage educational prototype addresses this gap by safely demonstrating pulse formation, energy transfer, and waveform control. This hands-on platform connects theory to measurable behavior while preparing students for careers in high-reliability research and national laboratory environments, and it is the main objective of this project.

Background

Pulsed power systems are widely used in modern electrical and industrial applications where high peak power is required over short durations. Unlike conventional continuous power supplies, pulsed power systems accumulate energy over time and release it rapidly, enabling high instantaneous power while maintaining relatively low average power consumption. This principle is fundamental to applications such as particle accelerators, pulsed radar systems, medical devices (e.g., defibrillators), and high-energy physics experiments [1], [2].

Several established pulsed power topologies exist in industry, each suited for specific voltage, current, and timing requirements. One of the most famous is the Marx generator, which charges capacitors in parallel and discharges them in series to achieve high voltage outputs [2]. Another common configuration is the Blumlein pulse forming network (PFN), which provides rectangular pulses with fast rise times and controlled impedance characteristics [2], [3], [4].

The system proposed in this project builds upon established pulsed power concepts while focusing on a low-voltage, microcontroller-controlled implementation suitable for educational use. The use of adjustable PWM control, current limiting, and real-time waveform visualization provides an educational platform for demonstrating switching behavior.

The novelty of this project lies in its combination of accessibility and functionality, offering a system that is both safe for laboratory environments and representative of real-world pulsed power systems. By incorporating both hardware and software elements, the design lets users explore the interaction between control algorithms and power electronics, providing valuable insight into modern electrical engineering practices.

Roadmap

This report first explains what the goals of this project are via sections like the Project Scope and Budget, which explain what the final design parameters are and what the Demo Day presentation will consist of. This report then lays out how the team has achieved this goal via the Project Setup and Completed Work listing out each major task necessary to have completed this project. The report will conclude by discussing the future scalability and potential market of the final prototype as well as the ABET Criterion taken into consideration throughout the process.

Project Scope

This project presents the design, construction, and validation of an adjustable pulsed DC power supply using PWM switching techniques. The hardware features N-channel power MOSFETs, isolated gate drivers, current-sense amplifiers, and microcontroller-based control enabling user-adjustable frequency and duty cycle. Physical implementation addresses breadboard layout for high-frequency switching, thermal management, and EMI mitigation. Experimental validation includes oscilloscope waveform characterization, efficiency analysis, and switching verification, demonstrating practical tradeoffs in a converter design suitable for motor control, LED driving, and battery charging applications.

A successful demonstration day presentation would be introducing the project objectives and final specifications (100Hz-8kHz frequency, 1-99% duty cycle, 1-12V

output, 0.6 W), then providing a hardware walkthrough identifying MOSFETs, gate drivers, microcontroller, inputs, and explaining the switching topology. The operating demonstration displays waveforms at various frequencies on the oscilloscope, showing voltage output at different duty cycles and presents timing measurements. Performance results include efficiency data across the operating range, thermal behavior under sustained operation, and comparison of measured versus theoretical values. A practical application demo shows LED dimming and blinking, followed by a brief conclusion summarizing achieved performance and design challenges overcome. The session concludes with Q&A where evaluators ask technical questions about design decisions and circuit behavior.

Budget Report

Total Estimated Project Cost: \$185.00

Category	Cost
Power Components (MOSFETs, gate drivers, capacitors, heatsinks)	\$68.00
Control Electronics (microcontroller, sensors, interface components)	\$42.00
Assembly Materials (board, terminals, wire)	\$35.00
Connectors & Hardware (jacks, enclosure, mounting)	\$18.00
Testing & Documentation (load resistors, thermal paste, printing)	\$22.00
Required Lab Equipment (Institution-Provided): Oscilloscope, multimeter, DC power supply (0-50V, 5A)	

Initial Setup

At the beginning of this project, before considering budget and safety limitations, the goal was to build an advanced and highly reliable PWM signal generator that could provide precise control over frequency, duty cycle, and output voltage while meeting and potentially surpassing the final target specifications of 100 Hz–8 kHz frequency, 1–99% duty cycle, 1–12 V output, and 0.6 W power delivery. Early research focused on understanding modern PWM techniques, efficient switching systems, and power electronics to explore the best possible design options. In an ideal scenario, the system would use a powerful microcontroller or FPGA for extremely accurate signal generation, dedicated gate drivers for faster and more efficient switching, and high-quality MOSFETs to improve performance and reduce power loss. Features like digital displays, software-based controls, and advanced testing equipment were also considered to make the system more

user-friendly and easier to analyze. Strong safety protections such as thermal monitoring, overcurrent protection, and voltage safeguards were part of the original vision as well. Although this setup would have created a more professional and flexible system, real-world limitations such as cost, complexity, and safety requirements made it necessary to scale the design into a simpler, more practical solution that still achieved the project's main objectives.

Work Completed

The main task first completed last semester is the STM32 microcontroller programming that controls the pulse width modulation. Controlling the PWM is an essential part of the project as it allows the user to change the duty cycle in real time, directly changing the average output voltage delivered to the load. The duty cycle is changed through a potentiometer and its current magnitude is represented with a blue LED for visual simplicity. This involved programming the STM32F407-Discovery board inside of its native software STM32CubeIDE by mapping the desired inputs/outputs for circuit components and implementing the correct functions in user code sections. As the team received the rest of the ordered parts, the PWM has been implemented with the gate driver that controls the switching of the MOSFET and drives the actual pulsing mechanism.

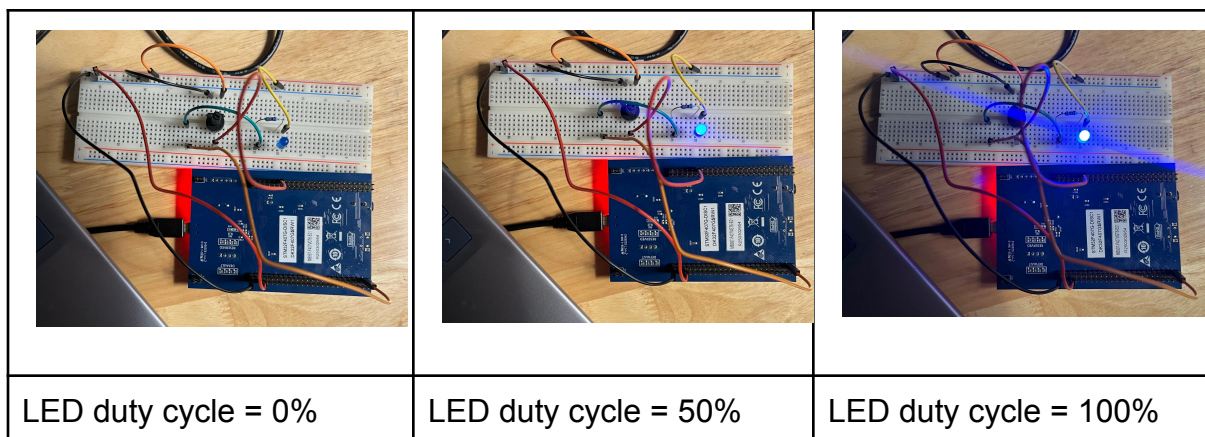


Figure 1: Microcontroller controlling blue LED duty cycle at various percentages in real time

The circuit shown above is simple and emits a constant voltage stream whose magnitude can be varied through the user input. This allows the team to demonstrate the overall concept of pulse power and changing the parameters results in a much more visible difference in output.

After completing the PWM, the team began assembling the IXDN602 gate driver and IRF540N MOSFET with the completed breadboard circuit. These two components are crucial to the overall function of the prototype, as the gate driver

controls the switching within the MOSFET, which creates the pulse train. This process involved remapping the inputs and outputs of the STM32 inside of STM32CubeMX, which allows for configuration of the clock cycle as well. The potentiometer remained to control the brightness of the LED, but the team decided to add a button as a separate input in order to toggle between two modes: pulse and constant voltage. This button when pressed would switch the potentiometer to vary the delay between pulses, allowing for control over the speed of the blinking LED. Results were verified through waveform visualization on an oscilloscope. Being able to control the pulse system powering a given machine is extremely important as many applications differ in pulse width, frequency, rise / fall time, and more. Controlling the frequency of the pulses through duty cycle control as well as brightness of the LEDs in two different modes demonstrates the general functionality of a pulse power controller.

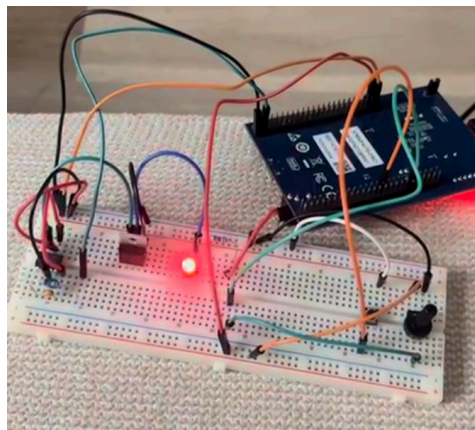


Figure 2: Prototype with gate driver and MOSFET driving LED pulsing

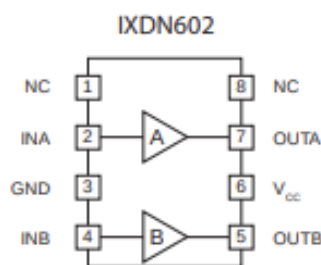


Figure 3: Pin configuration of IXDN602 gate driver

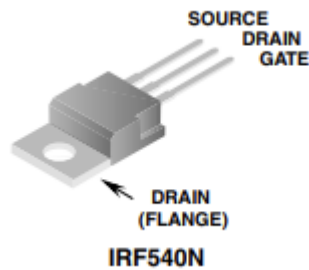


Figure 4: Pin configuration of IRF540N MOSFET

Future Goals

The project had to be scaled down for demonstration purposes, at 0.6 Watts. The 2.5 MW is obtainable with a higher budget and better resources. For future projects, it can be developed from the roadmap and schedule the team has created. An extension of the project can be done with more resources and a higher budget from a company that would be able to obtain it for themselves. This would extend the creation of the project by weeks and quite possibly months into the future.

Scaling it higher to the desired 2.5 MW would be a massive jump from where it is now, considering it would power a particle accelerator. Very achievable, but it will take time. Designing, maintaining, and troubleshooting megawatt-class modulators is impractical in typical student laboratories due to safety, cost, and facility constraints.

Potential Market

Organizations that design, operate, or maintain high-power pulsed systems (national labs, aerospace/defense contractors, specialized medical/industrial equipment makers). Primarily, military contractors would be most interested in pulse power systems technology with the rapid advancement of Artificial Intelligence. From being used in medical facilities to the chest-rigs on pilots, it is highly versatile.

ABET Considerations

Sustainability: Pulsed power technology is inherently linked to sustainability in the form of its energy usage. The alternative to supplying a quick pulse of voltage is through conventional DC or AC methods, which may not be necessary for all applications. Controlled bursts **reduce power consumption**, which can be particularly useful in machinery that require extremely high-voltage levels to run, such as a particle accelerator.

Manufacturability: Given the constraints of the supplies allocated to the group, the pulsed power display was **easier to manufacture** when compared to industrial applications. During the design process, the team focused on making an affordable demonstration without sacrificing quality of results. This pulse power supply is not meant for high-voltage but to demonstrate to a customer how this method works and how it can be implemented in certain machinery. The team's goal is to design a power supply not only easily comprehensible to a customer, but one that is easily reproducible as well.

Safety: The specific low-voltage topology in this project was chosen with the idea of customer safety in mind. Pulsed power modulation is much more common in high-voltage applications, but due to **safety concerns** commonly present in such situations, a low-voltage model only for demonstration will be used instead. The design utilized devices previously mentioned to ensure safety while testing, such as bleeder resistors and TVS diodes.

Conclusion

Team Pulse Power Innovators has successfully designed, tested, and implemented a pulse power supply prototype to a load for educational use. The final prototype, an estimated \$185, demonstrates the essential principle of a high voltage pulsed power supply safely through a breadboard-based circuit controlled by an STM32 microcontroller. The system has achieved a controllable duty cycle and frequency range within the specified project scope (1-99%), (100 Hz - 8 kHz) through a potentiometer and button. The actual pulse train is formed through solid state switching between the gate driver and MOSFET. This project has effectively bridged the gap between theoretical pulse power topology knowledge and actual hands-on experience designing, building, and implementing a scalable prototype demonstrating a technology necessary to many important industries.

References

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- [3] M. Reiser, *Theory and Design of Charged Particle Beams*, 2nd ed. Hoboken, NJ, USA: Wiley-VCH, 2008.
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